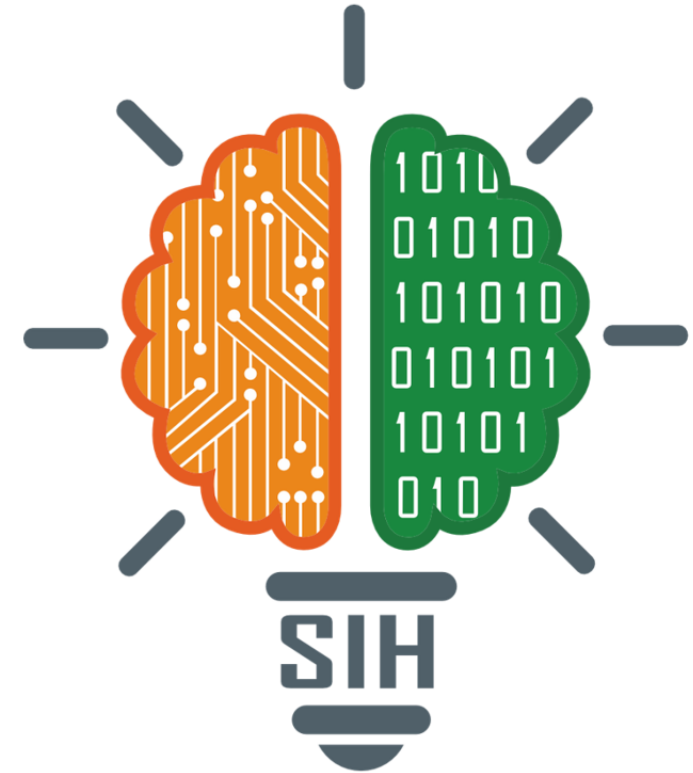


SMART INDIA HACKATHON 2026

AUTONOMOUS EXPLAINABLE TELE-SCREENING PLATFORM

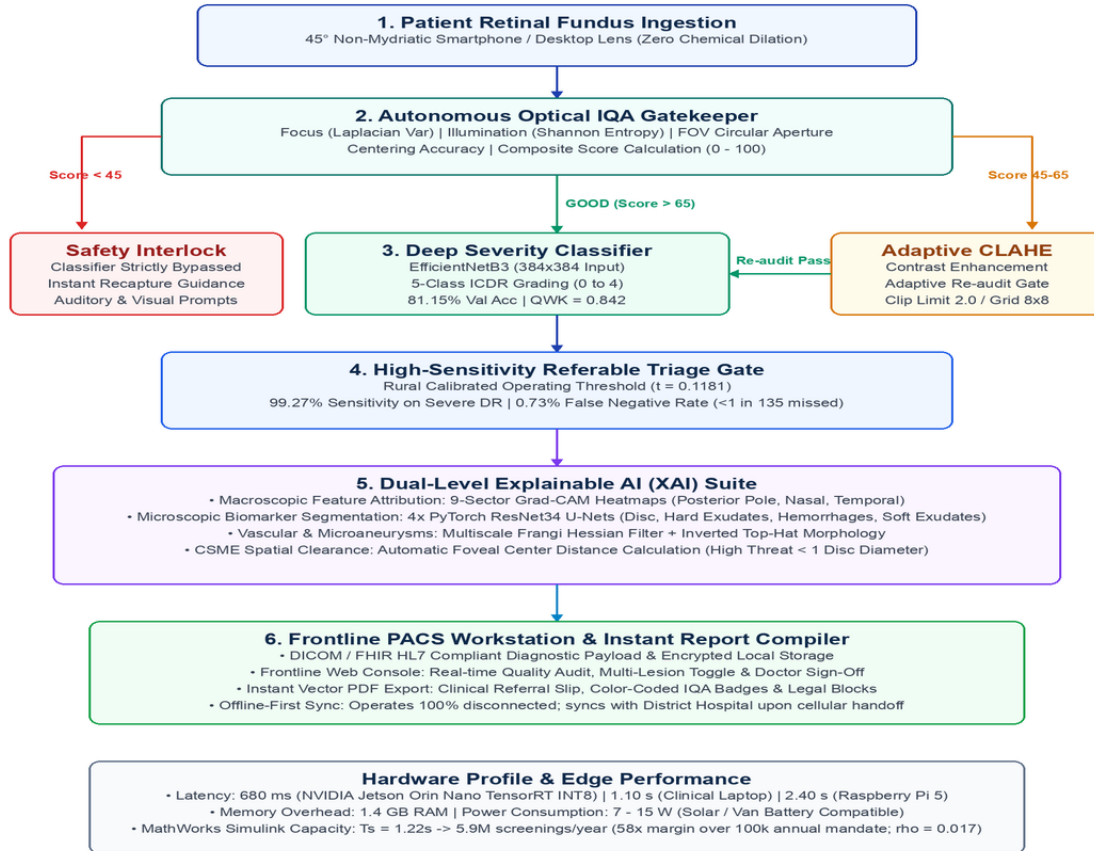
- **Problem Statement ID:** 26038
- **Problem Statement Title:** Explainable AI for Diabetic Retinopathy Screening in Rural India
- **Partner Organization:** MathWorks India
- **Theme:** MedTech / Healthcare & Biomedical Devices
- **PS Category:** Software
- **Team Name:** selfNprove
- **Team Leader:** Daksh Srivastava (dakshshrivastav56@gmail.com)
- **Team Members:** Samarth Navale, Aaryan Kuchekar, Gaurav Patel, Aaditya Bhosale, Jiya Jana
- **Registered Team ID:**



MathWorks Simulink 8-Subsystem Architecture Verified (DR_screening_workflow.slx)

8-SUBSYSTEM CLINICAL ARCHITECTURE PIPELINE

MathWorks Simulink Verified Screening Pipeline (DR_screening_workflow.slx)

**DEEP LEARNING & BIOMARKER METHODOLOGY**

- **Upstream Optical IQA:** 4-metric gate (Laplacian Focus + Shannon Illumination + Circular FOV + Centering). Ungradeable captures strictly bypass classifier.
- **5-Class ICDR Classifier:** EfficientNetB3 (384x384), 81.15% validation accuracy, Quadratic Weighted Kappa 0.842, cryptographically frozen checkpoint.
- **Dual Explainability Suite:** Macroscopic Grad-CAM heatmaps across 9 anatomical sectors + 4x PyTorch ResNet34 U-Nets (Optic Disc Dice 0.986, Exudates 0.758, Hemorrhages 0.748, Soft Exudates 0.760).
- **Vascular & Microaneurysms:** Multiscale Frangi Hessian filter (sigma 1.0-2.0) + inverted green-channel Top-Hat mathematical morphology (2-45 px pinpoint).
- **Referable Triage Gate:** High-sensitivity calibration (t = 0.1181) delivering 99.27% Sensitivity (0.73% False Negative Rate; <1 in 135 missed).

HARDWARE PROFILE & SIMULINK CAPACITY

- **Edge Inference Latency:** 680 ms on NVIDIA Jetson Orin Nano (TensorRT INT8) | 1.10 s on clinical laptop | 2.40 s on Raspberry Pi 5.
- **Memory & Power Profile:** 1.4 GB RAM overhead, 7–15W power consumption (battery/solar-van compatible).
- **Simulink M/M/1 Capacity:** Ts = 1.22s -> 5.9M screenings/yr capacity (58x margin over 100k annual mandate; rho = 0.017).

**TECHNOLOGY
STACK**

FastAPI


PyTorch


MathWorks


OpenCV


React


Docker


Keras


Python

ANALYSIS OF THE FEASIBILITY OF THE IDEA

- **Technical Feasibility:** Complete working prototype built and validated; frozen weights, 15 unit tests, 8 integration tests, 100% zero synthetic data.
- **Operational Feasibility:** Non-mydriatic handheld mobile fundus lenses (Remidio/Volk INR 15k–35k); ASHA workers operate after 15-minute standard training.
- **Financial & Social Feasibility:** Zero-cost village screening; saves rural families INR 2,000 in travel/wage loss while cutting government hospital crowding.

POTENTIAL CHALLENGES AND CLINICAL RISKS

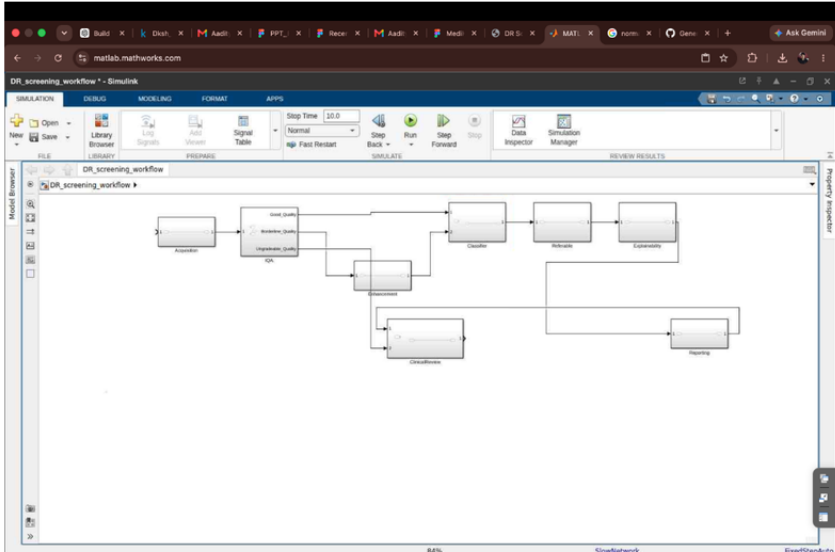
- **Optical Degradation in the Field:** Motion blur, eyelid tremors, and cataract opacity in rural elders cause severe fundus image artifacts.
- **Clinician Distrust of Black-Box AI:** Ophthalmologists legally reject AI predictions without visible, measurable anatomical proof.
- **Intermittent Rural Connectivity:** Primary Health Centres in remote tribal districts lack reliable broadband internet.

STRATEGIES FOR OVERCOMING CHALLENGES

- **Automated Hardware Safety Interlock:** Composite IQA score <45 strictly halts inference, giving instant voice/text guidance ('Steady lens / re-align pupil').
- **Epistemic Integrity & Biomarker Suite:** Every annotation is declared with provenance (GROUND_TRUTH vs DETECTED vs ESTIMATED) + CSME geometry.
- **100% Offline Edge Processing:** FastAPI + ONNX executes fully offline on local laptops; syncs encrypted reports only when network is present.

MATWORKS SIMULINK ARCHITECTURE & VERIFICATION

Formal 8-Subsystem Model (DR_screening_workflow.slx)



- **Strict Bypass Interlock:** Ungradeable quality automatically routes to recapture queue without executing neural weights.
- **Borderline Feedback Loop:** CLAE contrast enhancement dynamically evaluated before granting inference clearance.
- **M/M/1 Capacity Proof:** 5.9M peak screenings/year; steady-state queue length remains near zero over 100,000 requests.

FOR RURAL CITIZENS & FAMILIES

- **Vision Preservation:** Detects asymptomatic microvascular breakdown 12–24 months before vision impairment occurs.
- **Financial Protection:** Saves INR 1,500–3,000 in transit and 2 days of lost daily agricultural wages for 85% normal citizens.

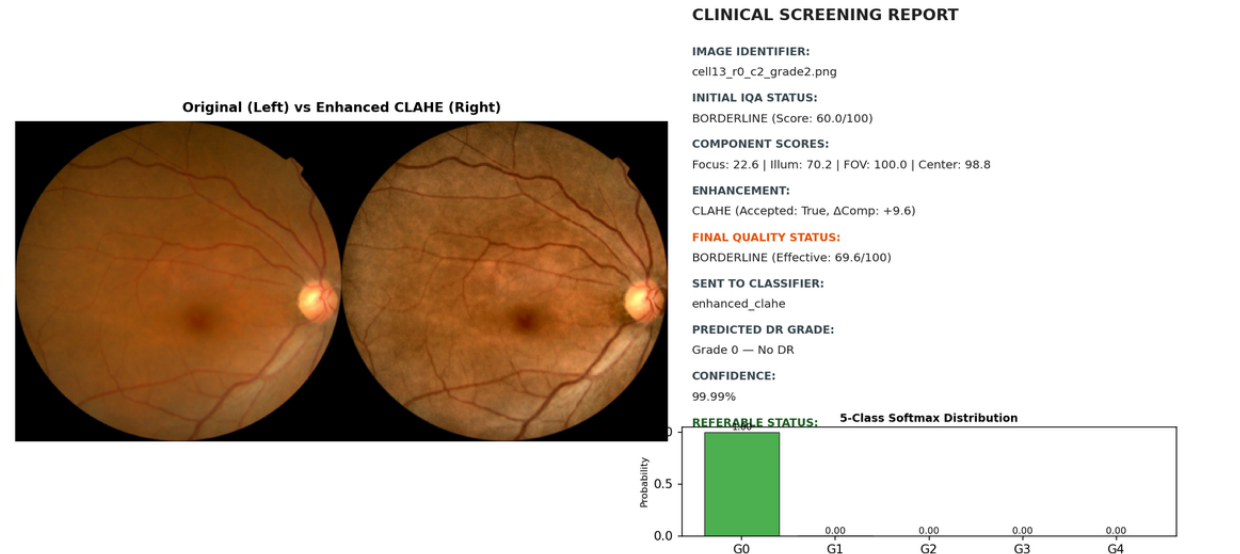
FOR FRONTLINE ASHA & ANM WORKERS

- **Zero-Risk Mobile Screening:** Non-mydriatic 15-second operation without chemical dilation drops or specialized expertise.
- **Instant Optical Feedback:** Automated focus/illumination prompts empower workers to capture clinical-grade fundus images.

FOR OPHTHALMOLOGISTS & PUBLIC HEALTH

- **District Hospital Decongestion:** Pre-filters healthy population, routing only high-priority referable cases (Grade 2–4).
- **Auditable Evidence:** Generates quantitative lesion contours, CSME clearance, and legal blocks in vector PDF reports.

CLINICAL USER STORY: SITAPUR PHC VALIDATION



- **Patient Scenario:** Ramesh Kumar (Age 58, Farmer) presented at village PHC with 10-year history of diabetes and zero visual symptoms. ASHA worker captured 45° non-mydriatic fundus in 15 seconds.
- **Automated Diagnostic Output:** IQA Score: 83.5 (GOOD) -> EfficientNetB3 classified Grade 2 Moderate NPDR (79.2% conf) -> Deep U-Net detected 46 Hard Exudate clusters within 0.8 DD of Fovea (HIGH CSME BLINDNESS THREAT).
- **Clinical Outcome:** High-priority referral triggered to District Eye Hospital within 14 days for anti-VEGF therapy. Irreversible macular edema prevented months before vision loss!

[1] IDRiD Benchmark Dataset

Indian Diabetic Retinopathy Image Dataset (IEEE Dataport)

Certified ground-truth pixel segmentations for microaneurysms, hemorrhages, hard/soft exudates, and optic discs.

[2] APTOS 2019 Blindness Detection

Asia Pacific Tele-Ophthalmology Society Dataset

Multicenter clinical fundus cohorts used for fine-tuning and evaluating the 5-class ICDR severity classification model.

[3] ICDR Clinical Grading Standards

International Council of Ophthalmology (ICO) Guidelines

Established the '4-2-1' clinical rule for Severe NPDR classification based on quadrant hemorrhage distributions.

[4] ETDRS Macular Edema Criteria

Early Treatment Diabetic Retinopathy Study Protocol

Geometric ground-truth standards governing Clinically Significant Macular Edema (CSME) based on foveal lipid distance.

[5] MathWorks Simulink Modeling

Discrete-Event Architecture & M/M/1 Queue Simulation

Verified 8-subsystem screening workflow (DR_screening_workflow.slx) establishing 5.9M annual throughput capacity.

[6] Frangi Multiscale Hessian Filter

IEEE Transactions on Medical Imaging (TMI)

Vesselness filter algorithm used to isolate retinal microvasculature calibers, arcade topology, and neovascular loops.

[7] Grad-CAM Explainability

Selvaraju et al., IEEE ICCV Framework

Convolutional feature attribution technique extracting gradient heatmaps across a 9-sector anatomical coordinate grid.

[8] EfficientNet & U-Net Architectures

Tan & Le (ICML 2019) | Ronneberger et al. (MICCAI 2015)

Foundational deep neural architectures powering 5-class severity classification and multi-lesion pixel segmentation.